AN2DL - Second Challenge Report
Coldplay

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December 16, 2025

1 Introduction

This report aims to provide a comprehensive analysis of the work conducted for the AN2DL second challenge. The project focuses on the classification of medical images showing diseased human tissues, to be classified based on cancer molecular subtype. For this purpose, different architectures of Convolutional Neural Networks have been explored, combined with domain-specific techniques for the image processing part.

2 Problem Analysis

2.1 Data characteristics

The dataset under examination comprises 691 images of different sizes coming from low-magnification WSIs of breast tissues, each paired with a binary mask highlighting most likely informative regions. Images are associated with one of four different labels to be predicted, identifying cancer molecular subtypes: *Luminal A*, *Luminal B*, *HER2(+)* and *Triple negative*.

2.2 Main challenges

The task we were assigned presented multiple difficulties. First, classes were not balanced in the dataset, as shown by the following distribution: *Luminal A* 30% of the samples, *Luminal B* 32%, *HER2(+)* 27% and *Triple negative*, the minority class, 11%.

In addition, all images had different rectangular shapes and were characterized by extended backgrounds, with informative tissues, the ones highlighted by the masks, often covering just small regions. Also, related works on histopathological WSIs (Whole Slide Images) typically rely on much higher-resolution images, that are able to significantly improve results.

Finally, the presence of artificially generated outliers, presenting green stains and extraneous patches, required additional efforts during the preprocessing phase.

3 Method

In this section we outline our approach, describing the rationale behind used methods and techniques. Given the unbalanced classes, the performance evaluation metric is the F1-score.

3.1 Data preprocessing

As first step, the outliers that were identified during the data exploration phase were removed from the dataset. We then processed images to make them of suitable shape for our networks. Since plain resizes

would have resulted in a loss of resolution and distorted aspect ratios, we opted for a **patch-based approach**. Exploiting mask information, we extracted patches around bounding boxes of mask connected components, to focus on more informative regions with reduced background, also increasing the number of samples. To avoid too overlapping patches, these were post-processed reducing their amount and maintaining sufficient information. An example of this is shown by Figure 1. After several trials, best results were obtained with 224x224 patches.

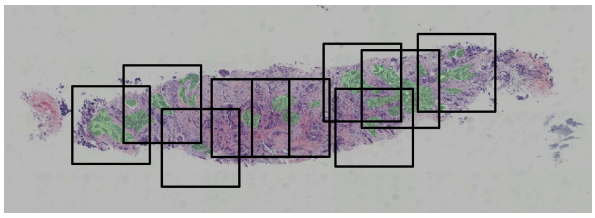

Figure 1: Sample image with extracted patches, mask regions in green.

Moreover, following an approach similar to the one proposed in [4], we also tried to extract patches of different sizes from the same mask regions (400x400, 200x200 and 100x100) and rescaling them to 100x100, simulating **different magnification levels** and at the same time giving more context, as depicted in Figure 2. The three collected images are then fed into three parallel CNN backbones and their extracted features concatenated before final classification.

To limit overfitting and improve generalization capabilities of our models, we constantly adopted **data augmentation** techniques at training time. Among all the explored ones, standard geometric augmentations yielded the best results, due to the nature of medical images.

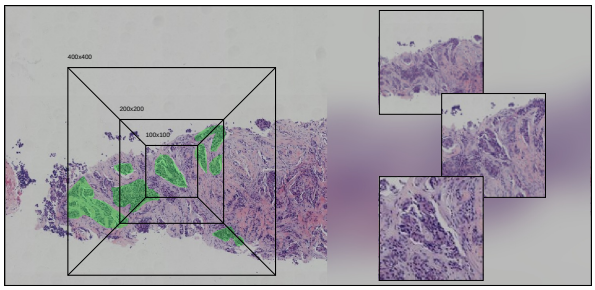

Figure 2: Multiscale images of a mask-highlighted region.

3.2 Model architectures

After experimenting with some under-performing custom CNNs, our first approach was based on **transfer learning** with different pre-trained FEN backbones, such as MobileNetV3 small, EfficientNetB3, ResNet-18 and ConvNeXt Tiny [3]. All of the mentioned architectures were trained to produce patch-level predictions and were later **fine-tuned** with smaller learning rates to adapt their final layers to our problem, increasing performance.

Afterwards, we employed some **model ensembling** techniques, with the aim of improving single architectures. First, latent representations extracted by different CNN backbones from the same patch were concatenated and fed into an MLP classifier, as proposed in [1]. Second, the multiscale approach was considered, as already introduced in the previous subsection, using the same FEN topology to process the differently-scaled versions of the same regions of interest.

Finally, after reviewing the literature to improve our results, we focused on **domain-specific architectures**. In particular, we made use of a ResNet-50 backbone, pre-trained with the RetCCL method [5], a clustering-guided contrastive learning technique designed for medical images. Furthermore, Prov-GigaPath, a visual transformer pre-trained on WSIs [6], was used as pure patch encoder, without further fine-tuning due to its size. This was combined with Attention-Based Multiple Instance Learning (ABMIL) [2]. Instead of directly classifying single patches, the model first projects the extracted features to a lower-dimension space and subsequently performs a weighted aggregation of the embeddings, to produce the global WSI prediction.

In addition to the cited techniques, we also tried to exploit binary masks feeding them as fourth channel or with a parallel CNN path, without noticing significant performance improvements.

3.3 Other techniques

In addition to architectural choices, all our models employed techniques to stabilize training and limit overfitting, such as early stopping, weight decay, dropout for classification layers and learning rate schedulers. Class imbalance was faced using a weighted cross entropy loss, based on class distributions. Furthermore different optimizers have been tested during the work, i.e. Adam, AdamW,

RAdam, Lion and Ranger. Finally, test-time inference was performed by averaging softmax probabilities over patches of the same image.

4 Experiments

In this section we describe the most significant experiments carried out during the project, among all the conducted ones. All results are reported in Table 1, with validation and test F1-scores. Although we present just single validation results, models were still tested on multiple splits for tuning and ongoing evaluation.

Single CNN backbones: first experiments involved pre-trained CNN models, then fine-tuned to adapt final layers to our data. ConvNeXt was the one achieving best results (Tiny version), combined with a single dense layer classifier of 512 neurons, layer norm and SiLU activation function. The model was trained using the Lion optimizer with rather small learning rates.

FEN ensembling: with the aim of enriching the latent representation, we combined three different pre-trained backbones, i.e. EfficientNetB3, ResNet-18 and MobileNetV3, exploiting the same classifier and optimizer described above. The resulting concatenated feature vector includes 2624 elements.

Multiscale ensembling: patches showing different scale levels of the same region were fed into three parallel ConvNeXt backbones, chosen as best-performing one among the tested CNNs. The concatenated feature vector includes 2304 elements. The selected classifier comprises one hidden layer of 256 neurons, with ReLU activation.

RetCCL ResNet-50: this was the first medical-specific FEN backbone, used to improve performance extracting more relevant features. The previous classifier was enlarged with an additional 1024-neurons layer. After unfreezing the last layer during the fine-tuning phase, this model achieved the best results on the test set.

Prov-GigaPath: this was the second experiment using a medical pre-trained architecture. Given the large number of hyperparameters (including the ones of the attention layer), tuning was performed with the *Optuna* framework. In the final configuration, the attention module has a 256-dimensional space, then classified by a 64-neurons hidden layer. This was also the model leading to best validation results.

5 Results and Discussion

Experiment	**Val-F1**	**Test-F1**
EfficientNetB3	0.4158	0.3854
ConvNeXt	0.4172	0.4281
FEN ensembling	0.4209	0.3929
Multiscale ensembling	0.4304	0.3709
RetCCL ResNet-50	0.4468	**0.4407**
Prov-GigaPath	**0.4961**	0.4279

Table 1: Validation and test F1-Scores.

As can be appreciated from Table 1, the usage of domain-specific architectures significantly enhanced results, regarding both validation and test performance. On the other hand, the usage of model ensembling techniques did not bring relevant improvements; in fact, these models were highly variable with respect to the given split, which also explains the substantial differences between validation and test scores.

In general, experiments did not yield outstanding results, even after fine-tuning was applied. A potential reason for this could be the fact that images at our disposal had a much lower resolution compared to the ones that are typically employed by studies of the field. Moreover, this probably reduces the effectiveness of the patch-based approach, in particular the usage of multiple-scale patches.

Finally, despite the usage of different data augmentation techniques, overfitting was still a relevant factor, likely due to the aforementioned reasons.

6 Conclusions

During this project, we faced the proposed image classification task exploring different methods and techniques, that were able to constantly improve our results. The adoption of models specifically trained on medical images, leading us to the best performance, is for sure a key aspect to highlight. Low magnification images were one of the main limitations of this work; therefore, possible future extensions might involve higher resolution WSIs, that already proved to be more promising in the existing literature. In addition, more advanced MIL approaches could be explored and integrated with the conducted experiments.

References

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